

Abdelrahman Wael Mohamed

Artificial Intelligence & Machine Learning Engineer

[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Phone Number & WhatsApp: +201029070837

Email: Abdelrahmanwael106@gmail.com

Cairo, Egypt

Summary

Artificial Intelligence graduate with expertise in machine learning, deep learning, natural language processing, computer vision, generative AI, Large Language Models (LLMs), and data science. Passionate about academic research, teaching, and developing intelligent systems that leverage AI to solve real-world problems.

Education

B.Sc. in Artificial Intelligence

Faculty of Informatics and Computer Science

The British University in Egypt (BUE)

2022 – 2026

Final Awarded Classification: **First Class**

Experience

Machine Learning Engineering Intern | DEPI Program

Oct 2024 - June 2025

- Completed an intensive Machine Learning Engineering internship with hands-on training in statistics, linear algebra, Python, data preprocessing, machine learning, deep learning, NLP, and computer vision.
- Trained and evaluated machine learning models for regression, classification, and clustering, focusing on performance optimization and accuracy improvement.

Generative AI Professional | DEPI Program

Nov 2025 – Jul 2026

- Completed intensive training in Generative AI, Large Language Models (LLMs), and Agentic AI systems.
 - Developed Retrieval-Augmented Generation (RAG) systems for knowledge retrieval and question answering.
 - Explored multi-agent architectures, AI deployment, monitoring, and responsible AI practices.
 - Worked with LLM APIs, prompt engineering techniques, and open-source AI models.
-

Projects

Hybrid-VTON: Diffusion-Based Virtual Try-On System (Graduation Project) ([View Project](#))

- Developed a Stable Diffusion XL-based virtual try-on framework supporting realistic upper and lower-body garment transfer while preserving human identity.
- Proposed a hybrid alignment framework combining explicit geometric matching with implicit correspondence learning, and designed a multi-conditioning architecture integrating IP-Adapter, ClothNet, EMASC, and dual GMMs to improve garment-body alignment while preserving garment details and semantic consistency.
- Trained and evaluated the framework on a unified VITON-HD + DressCode-MR dataset, achieving competitive performance across quantitative and qualitative virtual try-on benchmarks.

AI Fashion Designer: Multi-Agent Fashion Image Generation System ([View Project](#))

- Developed an end-to-end AI fashion design system integrating a conversational multi-agent framework with a fine-tuned Stable Diffusion XL (SDXL) model.
- Built an automated dataset preparation pipeline by generating image captions with Qwen2.5-VL and refining them through text preprocessing before fine-tuning SDXL.
- Designed a LangGraph-based multi-agent framework for clothing attribute extraction, conversational memory, prompt generation, and image generation workflow.

Sketch-to-Shoe Image Generation ([View Project](#))

- Developed a conditional latent diffusion model for generating realistic shoe images from edge sketches by extending the Diffusion Transformer (DiT) architecture with custom conditioning mechanisms.
- Designed and implemented custom conditioning mechanisms, including CLIP-based semantic conditioning and spatial latent conditioning by modifying the DiT input embedding layer.

Certificates

- AI & Data Science -Microsoft Machine Learning Engineer from DEPI.
 - Introduction to Machine Learning from AITB.
 - Supervised Machine Learning: Regression and Classification from DeepLearning.AI.
 - Advanced Learning Algorithms from DeepLearning.AI.
 - Python Programming fundamentals from Mahara Tech.
-

Skills

- **Programming Languages:** Python, C++, Java and SQL.
 - **Libraries & Frameworks & Tools:** NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Scikit-Learn, TensorFlow, PyTorch, and LangGraph.
 - **Machine Learning & AI:** Supervised and Unsupervised learning, Deep Learning, NLP, Image processing, Computer Vision, IoT, and Generative AI
 - **Soft Skills:** Communication, collaboration, problem-solving, teamwork, analytical mindset and prompt engineering.
-

Languages

- English (B2). - Arabic (Native)